

ANKIT KUMAR

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Education

Indian Institute of Technology, Kanpur

Master of Technology (M.Tech), Department of Management Sciences (DoMS)

Kanpur, India
July 2025 – Present • CPI: **7.18 / 10.0**

C.S.A. University of Agriculture and Technology

Bachelor of Technology (B.Tech) in Agricultural Engineering
Medalist

Kanpur, India
Aug 2021 – June 2025 • CPI: **8.25 / 10.0 (Chancellor's Silver Medalist)**

Professional Experience

Tata Consultancy Services (TCS)

Data Science & AI Intern | Computer Vision & Machine Learning

Remote / India
May 2026 – July 2026

- Engineered an automated vehicle CO₂ emission tracking pipeline from traffic CCTV feeds for enterprise ESG reporting.
- Implemented **YOLOv8** vehicle localization, **OpenCV** preprocessing, and **EasyOCR** with custom regex syntax correction.
- Trained dual-stage ensemble models: **Gradient Boosting** ($R^2 = 0.996$, RMSE = **3.12** g/km) and Random Forest; delivered interactive **Streamlit** dashboard with **17-test** **Pytest** validation suite.

Key Flagship Projects: Generative AI & Autonomous Agentic Systems

MediMind AI: Clinical Decision Support Platform

Stack: LangChain, Groq API (LLaMA-3.3-70B), Pinecone Serverless, MiniLM, Streamlit

Nov 2025 – Dec 2025
[\[GitHub Repo\]](#)

- Architected an end-to-end RAG diagnostic assistant using **LangChain**, **Groq LLaMA-3.3**, and **Pinecone**, cutting latency by **85%**.
- Integrated **MiniLM-L6-v2** dense vector embeddings, achieving **96.8% RAG faithfulness** with zero clinical hallucinations.
- Automated lab report ingestion via **PyPDF**, embedding 4 validated risk calculators with sub-10ms emergency triage guardrails.

Text-to-SQL & Agentic SQL Analytics System

Stack: LangGraph, Google Gemini 2.5 Flash, ReAct Agents, MySQL, RAGAS, Plotly

Jan 2026 – Feb 2026
[\[GitHub Repo\]](#)

- Developed an autonomous conversational database querying agent across a 6-table enterprise dataset with **64,000+ records**.
- Built an agentic **LangGraph ReAct** loop using **SQLDatabaseToolkit** with DDL/DML write-blocking security guardrails.
- Engineered self-correcting error handling resolving syntax anomalies iteratively, attaining **1.00 Context Precision** on **RAGAS**.

AI-Powered Gen Z Consumer Behavior & Marketing Platform

Stack: FastAPI, SHAP (Explainable AI), Unsupervised Clustering, Scikit-learn, Streamlit Cloud

Jan 2026 – Feb 2026
[\[Live App\]](#)

- Built an enterprise consumer intelligence platform integrating **SHAP explainability** and GenAI marketing copy generation.
- Architected high-throughput **FastAPI** inference backend for real-time persona scoring; deployed live on **Streamlit Cloud**.

Additional Relevant Generative AI & Agentic Systems (Full Portfolio)

- Agentic Multilingual Blog Generation & Fact Retrieval:** LangGraph, Groq LLaMA 3.1, DuckDuckGo Live Search API for citation verification.
- Multi-Agent Resume Analyzer & Job Matcher:** CrewAI 3-agent pipeline (Parser, Profiler, Recruiter Matcher) with cosine ATS scoring.
- Dental Clinic Appointment Booking System:** Stateful LangGraph graph, MySQL relational scheduling, conversational memory checkpoints.
- Air India Domain-Specific Knowledge Assistant:** LangChain, FAISS semantic vector search over airline regulatory transit manuals.
- Multi-PDF Conversational RAG Engine:** LangChain recursive text chunking, FAISS vector index, OpenAI LLMs with source citations.
- AI-Powered Sustainability & ESG RAG Chatbot:** Vector search over corporate environmental disclosure frameworks and compliance rules.
- Academic Research Literature RAG Assistant:** Semantic vector search over arXiv preprints with technical citation graph synthesis.

Technical Skills, Coursework & Certifications

- GenAI & LLMops:** LangChain, LangGraph, LlamaIndex, CrewAI, Groq API, Pinecone, FAISS, Qdrant, RAGAS, ReAct Agents, Prompt Engineering
- ML, DL & Frameworks:** PyTorch, TensorFlow, OpenCV, YOLOv8, Scikit-learn, Hugging Face Transformers, FastAPI, Streamlit, Git, Docker, Pytest
- Relevant Courses (IIT Kanpur):** Applied Machine Learning, Data Mining & Knowledge Discovery, Computing, Probability & Statistics
- Honors & Certifications:** GATE 2025: **98.48 %ile** • Chancellor's Silver Medal • Univ of Tokyo GCI AI • Oracle Certified DS & AI